

Package #168

Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continuously releases a heparin-mimetic, non-binding surface to inhibit biofilm and heparin-binding thrombus on the catheter lumen

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'heparin-binding thrombus formation' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters develop heparin-binding thrombus despite antithrombotic luminal coatings?". [fda_maude] 100 records retrieved live (1992-2070). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 1000003: Other [CARMEDA CIRCUIT - HEPARIN COATED]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Antithrombotic Polymers: A Narrative Review on Current and Future Strategies for Their Design, Synthesis, and Application.'; 'An anticoagulant activity and protein interaction study of chondroitin sulfate-coated magnetic nanocomposites.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Biofilm formation on catheter surfaces leading to protein adhesion and thrombus

Contents:

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Status: ENGINEERING_DEFINITION. Not physically validated.